R Programming Tutorial

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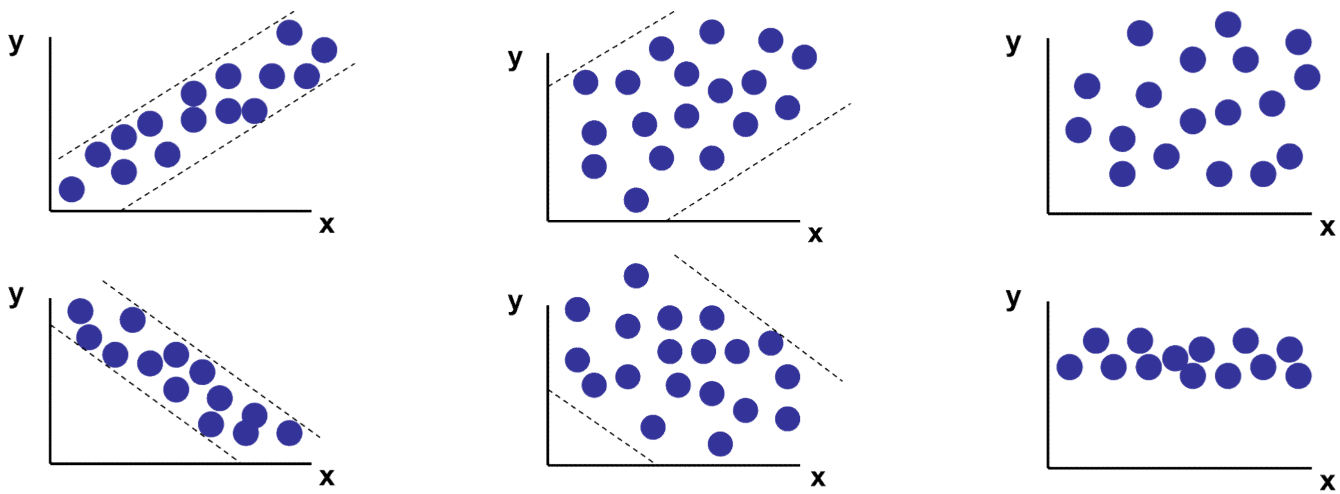
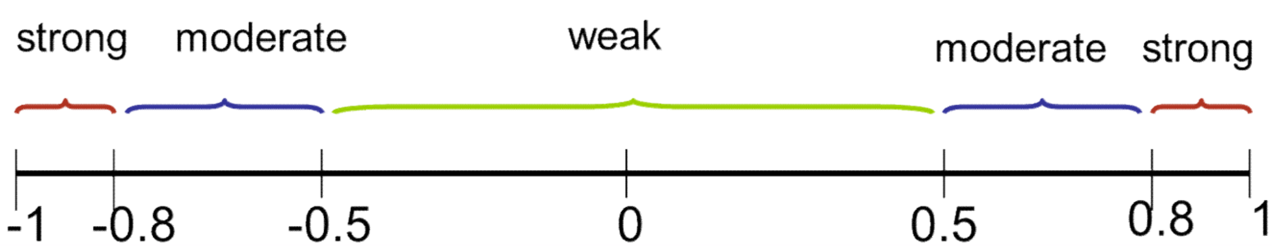
Correlation Analysis

- Measure the **strength** of relationship between **two** variables.



Strong relationship	Weak relationship	No relationship

Strength of correlation coefficient (r)



Correlation analysis in R

- Correlation efficient (r) values can be calculated in R via **cor()** function.
- How to use? **cor(x,y)**
  - The default method used is **Pearson's**
- Example data:

Students	Study Hours	GPA
S1	42	3.3
S2	23	2.9
S3	31	3.2
S4	35	3.2
S5	16	1.9
S6	26	2.4
S7	39	3.7
S8	19	2.5

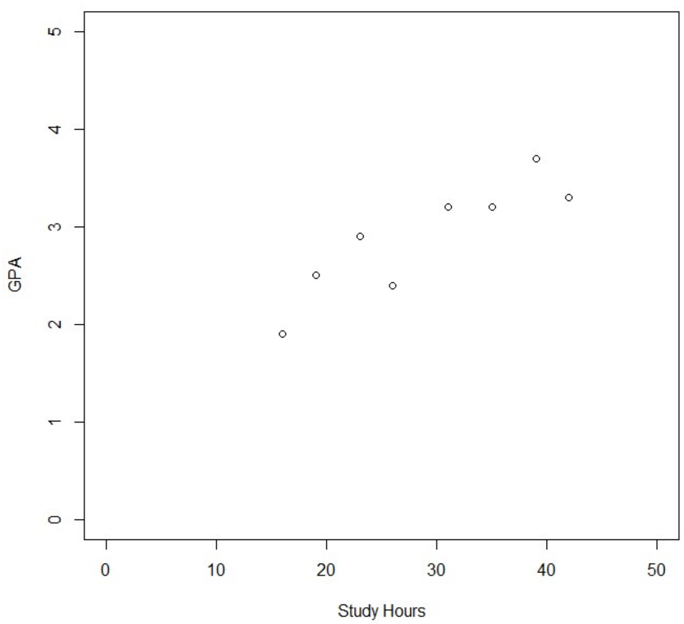
```
x <- c(42,23,31,35,16,26,39,19)
y <- c(3.3, 2.9, 3.2, 3.2, 1.9,
2.4, 3.7, 2.5)

    #calculate corr. coefficient
    cor(x,y)
```

```
> cor(x,y)
[1] 0.8837091
```

Correlation analysis in R

- How we can view the data in graph? via **Scatter plot**
- Use simple **plot()** function in R.




```
x <- c(42,23,31,35,16,26,39,19)
y <- c(3.3, 2.9, 3.2, 3.2, 1.9,
2.4, 3.7, 2.5)

    #calculate corr. coefficient
    cor(x,y)

    plot(x,y, xlim=c(0,50),
    ylim=c(0,5), xlab="Study
Hours", ylab="GPA")
```

Correlation analysis in R

- More example:

x	23	14	14	0	17	20	20	15	21
y	43	59	48	77	50	52	46	51	51

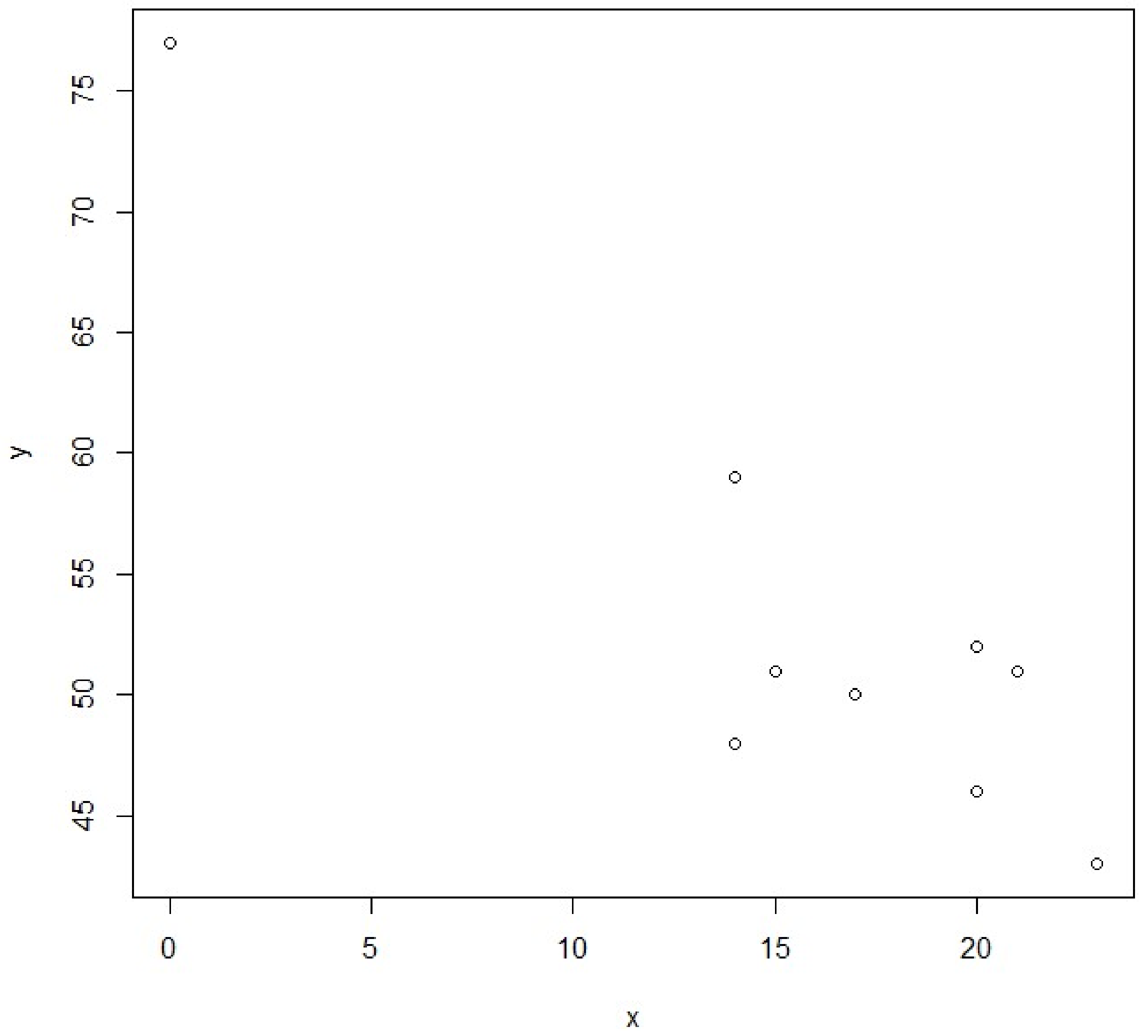
The data represent x=score on a measure of test anxiety and y=exam score for a sample of n=9 students:

a) Construct a scatter plot, and comment on the features of the plot.
b) Does there appear to be a linear relationship between the two variables? How would you characterize the relationship?
c) Compute the value of the correlation coefficient. Is the value of r consistent with your answer to part (b)?
d) Is it reasonable to conclude that test anxiety caused poor exam performance? Explain.

```
x<- c(23,14,14,0,17,20,20,15,21)
y<- c(43,59,48,77,50,52,46,51,51)

plot(x,y)
```

Correlation analysis in R



Correlation analysis in R

- More example:

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```
x<- c(23,14,14,0,17,20,20,15,21)
y<- c(43,59,48,77,50,52,46,51,51)

plot(x,y)

cor(x,y)
```

```
> cor(x,y)
[1] -0.9124331
```

Correlation analysis in R

- To perform linear regression in R, use the **lm()** function
- This will build a **linear regression model** based on the data
- How to use the function:
  - Assume you have your x, y
  - Use: **lm(y~x)**
  - Expression of "~" in R means it is a formula. Where y~x is equivalent to y = f(x)

```
x <- c(1400, 1600, 1700, 1875,
1100, 1550, 2350, 2450, 1425,
1700)

y <- c(245, 312, 279, 308, 199,
219, 405, 324, 319, 255)

model <- lm(y~x)
model
```

```
> model

Call:
lm(formula = y ~ x)

Coefficients:
(Intercept)            x
    98.2483       0.1098
```

Linear regression model

$$\hat{y} = 98.2483 + 0.1098x$$

Correlation analysis in R

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y <- c(245, 312, 279, 308, 199,
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model <- lm(y~x)
model

summary(model)
```

Linear regression model

$$\hat{y} = 98.2483 + 0.1098x$$

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Regression in R

Try adjust the plot with custom limits for x-axis and y-axis

$$\hat{y} = 98.2483 + 0.1098x$$

```
> summary(model)

Call:
lm(formula = y ~ x)

Residuals:
    Min      1Q  Median      3Q     Max 
-49.388 -27.388  -6.388  29.577  64.333 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)  
(Intercept) 98.24833   58.03348   1.693   0.1289  
x            0.10977    0.03297   3.329   0.0104 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 41.33 on 8 degrees of freedom
Multiple R-squared:  0.5808,	Adjusted R-squared:  0.5284 
F-statistic: 11.08 on 1 and 8 DF,  p-value: 0.01039
```

Correlation analysis in R

- What if we wan to see the plot of the model we learn from the data?
- First, plot a scatter plot based on x and y using **plot()** function
- Add the linear regression model into the plot using **abline()** function

```
x <- c(1400, 1600, 1700, 1875,
1100, 1550, 2350, 2450, 1425,
1700)

y <- c(245, 312, 279, 308, 199,
219, 405, 324, 319, 255)

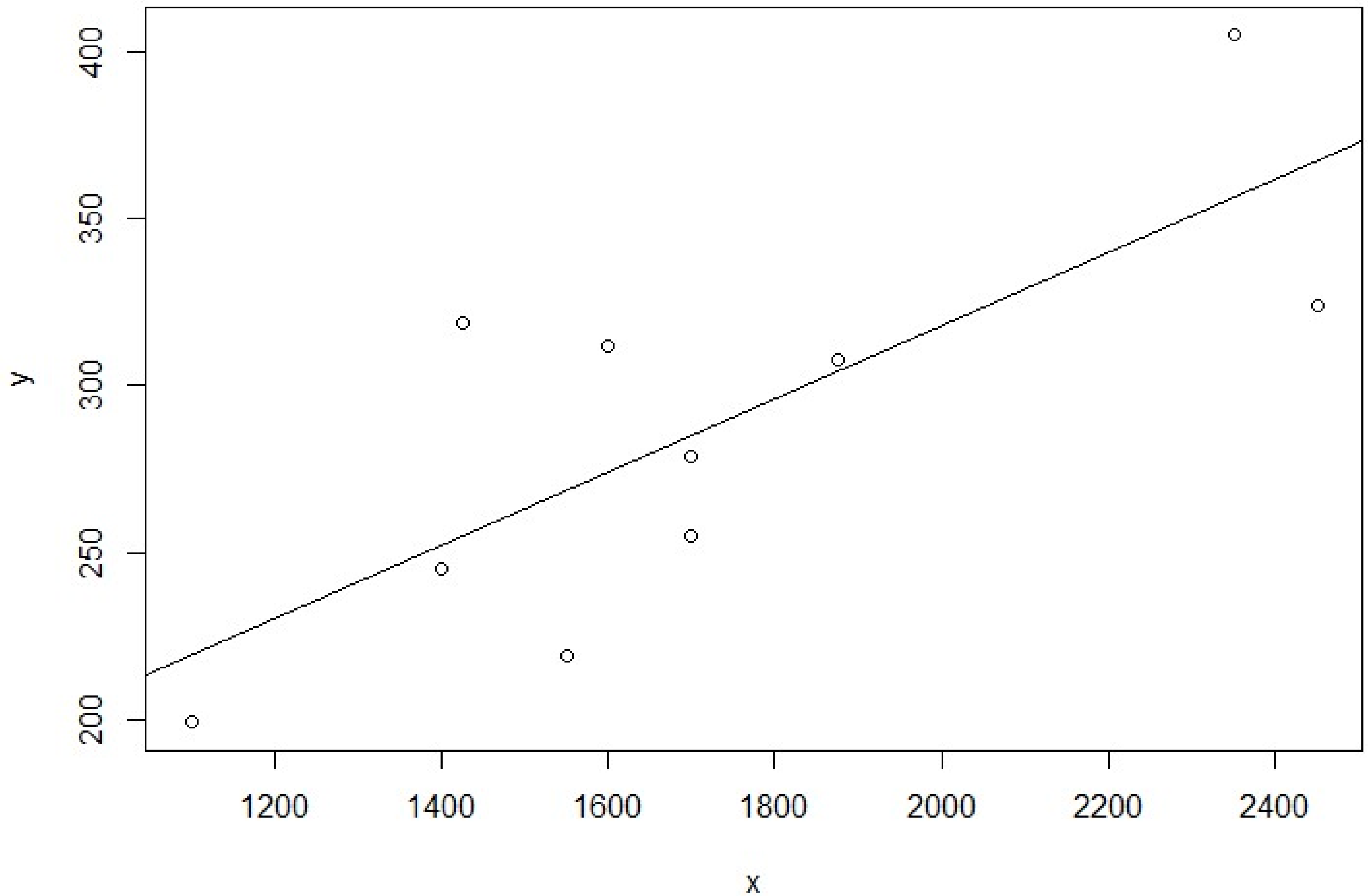
model <- lm(y~x)
model

plot(x,y)
abline(model)
```

Linear regression model $\hat{y} = 98.2483 + 0.1098x$

Regression in R

Try adjust the plot with custom limits for x-axis and y-axis



Regression in R

Replotting using:

```
plot(x,y, xlim=c(0,2600), ylim=c(0,500))
abline(model)
```